

2024 IEEE Open Conference of Electrical, Electronic and Information Sciences (eStream) | 979-8-3503-5241-2/24/\$31.00 ©2024 IEEE | DOI: 10.1109/ESTREAM61684.2024.10542593

3rdZeynep Hilal Kilimci

zeynep.kilimci@kocaeli.edu.tr

1) Our study makes significant contributions to the existing literature by presenting a comprehensive comparison of

In the hotel industry, customers express their experiences and rate hotels on various online platforms such as Book-

different neural network techniques in the context of sentiment classification for hotel reviews.

- 2) This study rigorously evaluates and compares the performance of various machine learning and deep learning models on IMDb datasets. This includes metrics such as accuracy, precision, recall, F1 score, and possibly others. The findings contribute to the understanding of the strengths and weaknesses of each model in sentiment analysis tasks.

This study comprises several sections. The first section introduces pertinent research progress in text sentiment analysis. Section III furnishes the details regarding the utilized models. The proposed framework is delineated in Section IV, and Section V delves into the experimental results associated with this framework. The study is summarized in Section VI, encapsulating the concluding remarks.

II. RELATED WORK

In this section, sentiment analysis-based literature studies are introduced on hotel reviews.

In [7], an extensive benchmark comparison of various deep learning architectures, including convolutional neural networks (CNN), long short-term memory (LSTM), and bidirectional encoder representations from transformers (BERT) with a Bi-LSTM, was conducted for sentiment analysis of drug reviews. This study explores diverse model combinations and assesses the impact of different pre-trained words embedding models. The results demonstrate that BERT consistently achieves superior performance but at the expense of significantly higher training time. CNN emerges as a time-efficient alternative with acceptable results, especially when initialized with static word embeddings, despite lower performance in classes with fewer training instances. These findings contribute valuable insights to the ongoing discourse on the selection of deep learning models for sentiment analysis in the domain of drug reviews.

In [8], sentiment analysis experiments were conducted on the Lithuanian Internet comment dataset, employing both traditional machine learning and deep learning approaches. The traditional machine learning methods, naive Bayes multinomial (NBM) and support vector machine (SVM), along with deep learning models, long short-term memory (LSTM) and convolutional neural network (CNN), were exploited to address the positive/negative/neutral sentiment classification task on both balanced and full dataset versions. The most promising deep learning outcomes, with an accuracy of 0.706, were observed when CNN was applied on top of FastText embeddings, emoticons were replaced, and diacritics were eliminated on the full dataset. Interestingly, traditional machine learning approaches, particularly NBM with replaced emoticons, restored diacritics, and lemma unigrams as features, demonstrated the highest accuracy of 0.735 on the full dataset. While traditional machine learning methods outperformed deep learning overall, the latter exhibited commendable results, especially on smaller datasets, indicating its potential in sentiment analysis tasks.

Muhammad et al. [9] addressed the challenge faced by online travel agents (OTAs) with a large volume of reviews by de-

veloping a sentiment assessment technique for hotel reviews in Indonesian. Using the LSTM and Word2Vec models, various parameters were explored, including Word2Vec architecture, vector dimension, evaluation method, LSTM dropout value, pooling technique, and learning rate. Through experimentation with a dataset comprising 2,500 review texts, optimal performance was achieved, attaining an accuracy of 85.96%. The most effective parameter combinations for Word2Vec were found to be the Skip-gram architecture, the Hierarchical Softmax evaluation method, and a vector dimension of 300. For LSTM, the successful parameter combinations included a dropout value of 0.2, average pooling as the pooling type, and a learning rate of 0.001. This study provides valuable insights for OTAs seeking to enhance their understanding of customer sentiments from several reviews.

In [10], a novel framework is introduced for evaluating online review ratings using machine learning techniques. This study investigates the performance of single and ensemble classifiers in predicting sentiment on diverse datasets, emphasizing the often-overlooked aspect of neutral polarity. This research refines classifier performance through varied pre-processing and feature extraction methods. Key findings highlight the effectiveness of the linear-kernel support vector machine, logistic regression, and multilayer perceptron as top classifiers, which are further improved in ensemble models. Text pre-processing techniques and sentiment lexicons contribute to increased accuracy. This research underscores the framework's reliability across different datasets, particularly in enhancing neutral sentiment accuracy without compromising positive or negative rating precision.

In the research conducted by [11], a combination of computational linguistics, visual analytics, and deep learning techniques was applied to sense hotel reviews and corresponding responses from TripAdvisor. The primary objective was to identify response strategies. A thorough examination of 113,685 hotel reviews and responses was carried out, encompassing their semantic and syntactic relationships. Pioneering the utilization of visual analytics and deep learning-based natural language processing, the study empirically identified managerial responses. The outcomes underscored the effectiveness of the proposed multi-feature fusion convolutional neural network model, showcasing its capacity to improve data complementarity and surpass comparative approaches. Visualization results not only contributed to improving the model's performance but also offered valuable insights into response strategies. This study makes significant theoretical and technical contributions by using innovative methodologies in the analysis of online hotel reviews and managerial responses.

In [12], neural networks were employed to extract opinions from movie reviews. The research used the "Movie Review Database by Stanford University" for training in conjunction with extensive lists of positive and negative words. The trained system demonstrated exceptional precision in its results. The neural network model, consisting of six layers (four hidden, one input, and one output) with a single neuron for binary classification (positive or negative review), was specifically

designed for movie reviews. Remarkably, the training accuracy of the model reached 91.9%, and the validation accuracy stood at an impressive 86.67%. This showcases the effectiveness of the neural network-based approach in accurately classifying sentiments in movie reviews.

III. METHODS

In this section, we present deep learning methods created specifically for conducting sentiment analysis.

A. Convolutional Neural Network (CNN)

CNN finds extensive applications across various domains, demonstrating remarkable performance in tasks such as image recognition, face recognition, and handwritten digit recognition. The key components of the CNN include convolutional layers and pooling layers. These networks are characterized by local vision sharing, weight sharing, and pooling, which significantly reduce model parameters and accelerate training. The advantage of weight sharing lies in the independence of feature extraction parameters during convolution, as the same parameters are applied uniformly across positions [13].

In the convolution layer, an $n \times n$ convolution kernel convolves information from input layers, employing multiple convolution kernels to form the layer. The CNN structure comprises three primary layers. The initial layer, convolution, convolves the input matrix with kernels to identify specific features. Numerous filters, initialized with a Glorot uniform distribution, undergo training adjustments to enhance feature detection. The second layer, pooling, combines neuron cluster outputs into a single neuron in the next layer through processes such as Max Pooling or Average Pooling, often favoring Max Pooling for its common usage. This layer introduces translation invariance. Situated between two convolution layers, the pooling layer facilitates seamless information flow.

In the layer lineup, the third layer is called “fully connected,” and it comes after the convolution and pooling layers, playing a big role in how the network works. The information gathered from the convolutional and pooling layers goes into this fully connected layer, which then passes the recognized features into a hidden layer used for classification. Because we are dealing with a task that involves sorting things into two categories, the output layer is set up with just one point and uses a sigmoid activation method. We use two convolutional layers and two pooling layers to avoid the complexity of the model.

B. Long Short Term Memory (LSTM)

LSTM is designed to handle long-term dependencies more effectively. Like RNN, LSTM also has a temporal loop within its layer, but the key distinction lies in the memory cell. This memory cell can either store or update information based on the input sentences provided. Within an LSTM cell, there are three main gates. The first is the Forget Gate layer, which acts like a filter to decide what information should be discarded from the memory cell. The second layer is the input gate layer, which is responsible for determining what new information

should be stored in the memory cell. This layer consists of two parts: a sigmoid layer and a tanh layer. The third gate is the output gate layer, which determines the final output of the respective LSTM cell [14].

In our model, we use the same input for both the LSTM and CNN networks. In the LSTM network, we keep things simple by using just one LSTM layer after the Embedding Layer. This helps avoid making the model too complicated while still allowing it to work well for sequential data. The output layer is identical to that of CNN.

C. CNN-LSTM

A CNN-LSTM model is a hybrid neural network architecture that combines the strengths of CNNs and LSTMs. This model is often used for tasks involving sequential data, such as time series analysis, video processing, and natural language processing [15].

In our CNN-LSTM design, we have merged two models to enhance our approach. Initially, we employed a trained specialized embedding layer to embed our input sentence. Next, the input was processed through a two-layer convolution network. The result of the maxpool layer was then directed into the LSTM layer, which plays a role in analyzing the order of the text. The outputs from the LSTM layer were further fed into a fully connected layer. This layer transfers the identified features to a hidden layer designated for classification purposes. This integrated architecture effectively captures and analyzes sequential information within the input text for improved text classification outcomes.

D. Bidirectional LSTM (BiLSTM)

Bidirectional LSTM (BiLSTM) [16] is a type of recurrent neural network commonly applied in natural language processing. Unlike the standard LSTM, this model processes input in both forward and backward directions, enabling it to incorporate information from both ends of the sequence. It proves to be a robust tool for modeling sequential relationships among words and phrases bidirectional. To simulate the physical layout of equipment deployment, BiLSTM networks are employed to capture spatial information from the entire reactor network at a given moment. Subsequently, a unidirectional LSTM layer is employed to extract temporal features from multiple time steps. The predictions suggest that using deep learning for feature extraction is advantageous in capturing the nonlinear and dynamic characteristics of complex reactor systems, contributing to more accurate predictions for soft sensing applications.

In our approach, we maintain simplicity by incorporating only a single BiLSTM layer following the embedding layer. This streamlined design ensures that the model remains straightforward, preventing unnecessary complexity while retaining effectiveness in handling sequential data.

E. Convolutional BiLSTM (Conv-BiLSTM)

Conv-BiLSTM is a distinct method for constructing a model, particularly applied in natural language processing

(NLP). This design blends two robust techniques: convolutional neural networks (CNN) and bidirectional long short-term memory (BiLSTM). The objective is to grasp features at both the character and word levels, thereby enhancing the model's capacity for making precise predictions in NLP tasks. This innovative combination leverages the strengths of both CNN and BiLSTM for improved performance in understanding and predicting natural language patterns [17].

In our Conv-BiLSTM approach, we started by embedding our input sentence using a dedicated embedding layer that underwent training. Afterward, the input was processed through a single-layer convolution network. The outcomes of the maxpool layer were then directed into the BiLSTM layer. Subsequently, the results from the BiLSTM layer were further transmitted into a fully connected layer to obtain the final classification result. This integrated method captures and understands features within the input data effectively to improve classification outcomes.

IV. PROPOSED FRAMEWORK

In this section, we provide an overview of the datasets utilized in this investigation and introduce the suggested methodology. The dataset for this study is a dataset of hotel reviews of four hotels in Indonesia obtained from two of the most popular online travel platforms, namely TripAdvisor and Hotels.com. The extensive and abundant content generated by users on both platforms significantly increases the likelihood of gathering comprehensive and sufficient data for analysis. Hardrock, Padma, Ayana, and Pullman Hotel were selected for this study because each had over 5,000 reviews, making them the largest in quantity compared to other hotels.

Customer review data retrieval occurs through web crawling using WebHarvy. The raw dataset consists of 20,000 reviews from four hotel reviews. Before the analysis of the proposed system, the following data pre-processing steps are ensured to eliminate inconsistencies by cleaning the reviews in the collected dataset:

- 1) If a review is not written in English, then it is discarded.
- 2) Neutral comments, if any, are not included.
- 3) Punctuation marks, numbers, and emojis are removed from the reviews.

The Python library Natural Language Toolkit (NLTK) is used for natural language processing to clean the collected data by eliminating unnecessary information. The word removal function is applied to filter out common words such as "is", "the", "and", "a", etc., which are deemed to have minimal significance in text analysis. The lowercase function is employed to convert all letters in the text to lowercase. Finally, the punctuation removal function is used to eliminate all punctuation marks from the text. Some examples of the use of data pre-processing techniques performed on hotel reviews in the dataset are illustrated in Table I.

The sensitivity polarity feature of the TextBlob library in Python is used to label hotel reviews as positive or negative. TextBlob has predefined rules with certain scores to aid in calculating the polarity of a sentence. TextBlob returns a value

TABLE I: Example of data pre-processing for hotel reviews

Reviewed Text	Preprocessed Text
Padma resort ubud offers you the best experience with all the great experiences and qualities services, they are trying to personalized the service which are so great so the guests feel welcomed and more enjoying the staynces and qualities services	"padma", "resort", "ubud", "offers", "best", "experience", "great", "experiences", "quality", "services", "strive", "personalize", "service", "making", "guests", "feel", "welcomed", "enjoy", "stay"
We currently stayed at Padma room. The big was spacious with bath tub. However, we are quite disappointed with room house-keeping. It was quite poor house-keeping especially hygiene.	currently", "stayed", "padma", "room", "big", "spacious", "bath", "tub", "quite", "disappointed", "room", "housekeeping", "quite", "poor", "housekeeping", "especially", "hygiene"
Hard Rock Hotel Bali is an excellent resort. It has amenities for all ages, and for all group types. The main pool is huge which means there is room for everyone. The food is great, the location is excellent, and the staff are wonderful.	"hard", "rock", "hotel", "bali", "excellent", "resort", "amenities", "ages", "group", "types", "main", "pool", "huge", "means", "room", "everyone", "food", "great", "location", "excellent", "staff", "wonderful"
Unfortunately the Hard Rock in Kuta was quite disappointing for us on our family holiday. As usual staff were excellent but they always are in Bali. The hotel was overpriced and the room quite dated.	"unfortunately", "hard", "rock", "kuta", "quite", "disappointing", "family", "holiday", "usual", "staff", "excellent", "always", "bali", "hotel", "overpriced", "room", "quite", "dated"

in the range of [-1, 0, 1]. In this study, data with values below 0 are labeled as having negative polarity, whereas data with values above 0 are labeled as having positive polarity. Table II shows the distribution of sentiments in the dataset after labeling.

TABLE II: Statistics of the datasets before the balancing process

Dataset	Positive	Negative	Total
Padma Hotel	4,798	202	5,000
Hard Rock Hotel	4,178	822	5,000
Ayana Hotel	4,385	615	5,000
Pullman Hotel	4,198	802	5,000

Upon labeling the data, we encounter the issue of class imbalance. To address this, we employ ChatGPT to rephrase the negatively tagged data, ensuring a balance with the positively labeled data. We ensured equal distribution within each dataset, resulting in 8,000 reviews per dataset, comprising 4,000 positively labeled and 4,000 negatively labeled reviews.

TABLE III: Detailed analysis of the dataset

Dataset	Numb of Class	Average Length	Max Length	Vocab Size	Train	Test
Padma	2	79	1,259	13,524	6,400	1,600
Hard Rock	2	72	1,323	13,223	6,400	1,600
Ayana	2	100	2,202	16,157	6,400	1,600
Pullman	2	83	1,604	13,824	6,400	1,600
IMDb [18]	2	231	2,470	141,122	25,000	25,000

Detailed analysis of the datasets is shown in Table III. We incorporate detailed information from IMDb into the table to compare the selected deep learning models with machine

learning models. We chose the IMDb dataset because it is commonly utilized in comparable studies, especially in the field of machine learning. Table IV presents the sentiment labels of some texts obtained after pre-processing.

TABLE IV: Sentiment labels on the dataset after pre-processing

Review	Sentiment
padma resort ubud offers best experience great experiences quality services strive personalize service making guests feel welcomed enjoy stay	Positive
currently stayed padma room big spacious bath tub quite disappointed room housekeeping quite poor housekeeping especially hygiene	Negative
hard rock hotel bali excellent resort amenities ages group types main pool huge means room everyone food great location excellent staff wonderful	Positive
unfortunately hard rock kuta quite disappointing family holiday usual staff excellent always bali hotel overpriced room quite dated	Negative

To achieve optimal performance in practice, the hotel review dataset is divided into two subsets for training and testing, with proportions of 80% and 20%, except IMDb dataset. The distribution of dataset sharing in the IMDb dataset aligned with established literature. The learning rate was set to $2e-5$, and the batch size was set to 32. The models are trained for 10 epochs. The same computing environment (Tesla T4 GPU) was used for the training process. Detailed hyperparameter settings for each model are provided in Table V.

TABLE V: Hyperparameters for our experiments

Learning Rate	Embedding Dimension	Batch Size	Max Length	Optimizer
$2e-5$	300	32	256	Adam

Table VI provides a detailed overview of the parameter counts used in each of the selected deep learning models. Understanding these parameter numbers is essential to gain insights into the complexity and computational needs of the chosen deep learning architectures.

TABLE VI: Number of parameters in each model

CNN	LSTM	CNN-LSTM	BiLSTM	Conv-BiLSTM
4,735,137	4,373,089	4,546,613	3,191,231	3,831,711

In this study, the proposed framework is designated to classify hotel reviews as positive or negative by employing deep learning models. Moreover, the performances of the models are compared on employed Padma, Hard Rock, Ayana, and Pullman hotel review datasets.

A general flowchart of the proposed framework is shown in Figure 1. Our models are built on the powerful and versatile Keras framework. With its intuitive and user-friendly interface,

Keras serves as the backbone for the implementation of diverse machine learning models [19]. Finally, the sentiment classification performance of the models was assessed using accuracy, precision, recall, F1 score, and AUC evaluation metrics.

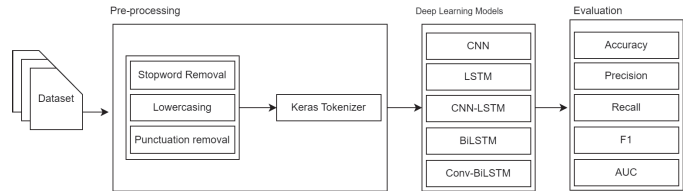


Fig. 1: A general flow chart of the proposed framework.

V. EXPERIMENT RESULTS

In this section, we present the results of experiments conducted on well-balanced datasets, showcasing the performance of various deep learning models. The classification performance is indicated by ACC, which serves as an abbreviation denoting accuracy score presented in percentages within the tables. The implementation of the models was executed on Google Colab, using its GPU resources freely. The experiments involved 80% training and 20% testing data separation, employing a repeated holdout technique conducted 10 times across the entire database. The notations used to represent the experimental techniques include CNN, LSTM, CNN-LSTM, BiLSTM, and Conv-BiLSTM. The highest accuracy scores are highlighted in bold.

First, we offer a comparative analysis of the experimental results on hotel review datasets using deep learning models, as illustrated in Table VII.

TABLE VII: Accuracy results of deep learning models on hotel review datasets

	Padma	Hard Rock	Ayana	Pullman
CNN	94.42	82.40	81.21	83.29
LSTM	96.20	85.31	85.23	85.14
CNN-LSTM	95.36	84.07	85.87	85.45
BiLSTM	95.12	83.93	84.90	86.09
Conv-BiLSTM	94.68	83.75	84.61	85.77

The analysis in Table VII clearly indicates that the LSTM model exhibits superior classification success compared with other deep learning models for two of four hotel review datasets. Specifically, LSTM outperforms the others, achieving an accuracy score of 96.42% on the Padma Hotel dataset and 85.31% on the Hard Rock Hotel dataset. Subsequently, CNN-LSTM demonstrated an accuracy of 95.36%, and BiLSTM achieved an accuracy of 95.12% on the Padma Hotel dataset. In addition, the CNN model shows the lowest performance with an accuracy of 94.42%, approximately 2% below the best-performing model, LSTM. In summary, the classification performance of each deep learning model, based on the accuracy metric, can be summarized as follows: LSTM > CNN-LSTM > BiLSTM > Conv-BiLSTM > CNN.

To validate the performance of the proposed framework, experiments were also conducted on widely used IMDb

datasets. The IMDb dataset revolves around a binary classification problem that distinguishes between positive and negative reviews. To assess and compare the performance of the machine learning and deep learning architectures, we employed various accuracy metrics derived from the confusion matrix. Table VIII presents a comparison of the proposed deep learning models with state-of-the-art studies that concentrate on different machine learning methods for IMDb dataset. The performance of each model was assessed using multiple evaluation metrics, including accuracy (Acc), precision (Prec), recall, F1 score (F1), and area under the curve (AUC). The F1 measure, which serves as the harmonic means of precision and recall, provides a balanced evaluation by considering the trade-off between false positives and false negatives in classification.

TABLE VIII: Comparison of the proposed models with machine learning methods on the IMDb dataset

Models	Acc	Prec	Recall	F1	AUC
Unigram+ME [20]	83.65	84.00	83.00	83.00	-
TriGram+NB [21]	79.16	79.00	79.00	79.00	-
TriGram+SVM [21]	73.36	74.00	73.00	73.00	-
TF-IDF + KNN [22]	50.80	50.00	72.00	59.00	-
TF-IDF + RF [22]	74.80	72.00	81.00	77.00	-
AdaBoost [23]	82.93	-	-	-	80.00
XGBoost [23]	82.93	-	-	-	81.00
CNN	87.48	87.25	86.86	87.31	87.19
LSTM	84.03	83.80	84.15	84.22	84.29
CNN-LSTM	82.55	82.26	82.44	82.10	82.14
BiLSTM	84.25	84.37	84.07	83.74	84.46
Conv-BiLSTM	78.79	78.92	78.56	79.04	79.32

It is clearly seen from the experiments that the highest values of evaluation metrics are obtained with CNN. CNN outperforms others by resulting in an accuracy of 87.48%. CNN that got the lowest score on the various hotel datasets showed different performance on the IMDb dataset. On the IMDb dataset, which has a larger size, CNN performance increased along with complexity. It is followed by BiLSTM with 84.25% accuracy and LSTM with 84.03% accuracy. Both showed similar performance, performing approximately 3% below compared with CNN.

Loss vs epoch size visuals are demonstrated in Figure 2, 3, 4, 5 and 6 to indicate how we handle the over-fitting problem. In this investigation, the best models are : the LSTM model on the Padma Hotel and Hard Rock dataset, the CNN-LSTM model on the Ayana Hotel dataset, the BiLSTM model on Pullman Hotel dataset and the CNN model on the IMDb dataset. In addition, we utilized the ROC curve to assess our classifier's performance. The ROC curve allows the results to be analyzed more insightfully, given the binary classification. In the current study, the top-performing models are as follows: The LSTM got the highest score of 96 on Padma Hotel dataset and 85 on Hard Rock Hotel dataset. CNN-LSTM scored 86 on Ayana Hotel dataset, and BiLSTM got 86 on Pullman Hotel dataset as illustrated in Figure 7, 8, 9 and 10. The top-performing models on IMDb dataset is CNN. CNN model obtained a score of 87 as illustrated in Figure 11.

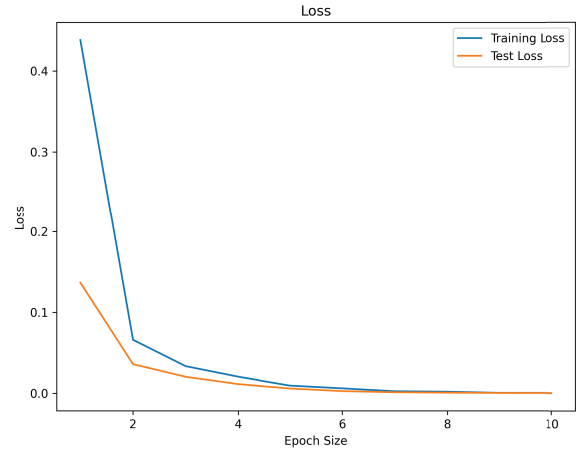


Fig. 2: Loss graphics of LSTM on Padma dataset

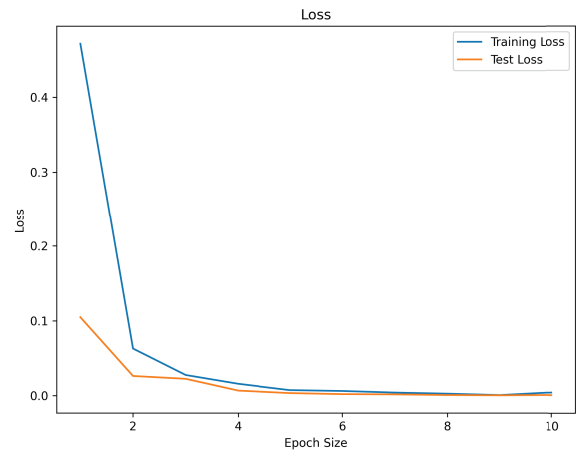


Fig. 3: Loss graphics of LSTM on Hardrock dataset

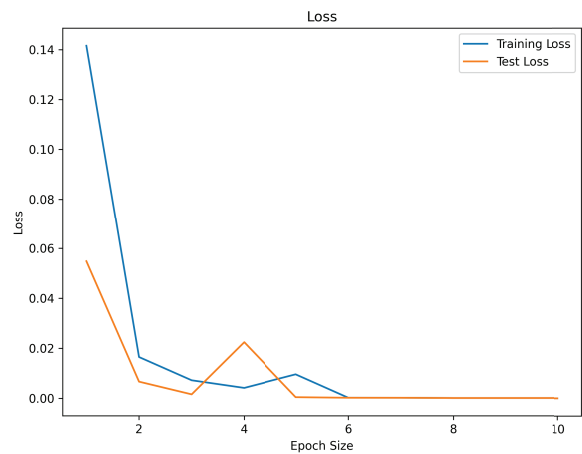


Fig. 4: Loss graphics of CNN-LSTM on Ayana dataset

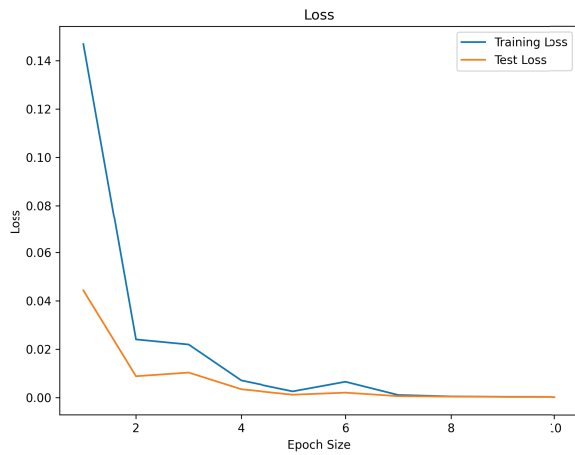


Fig. 5: Loss graphics of BiLSTM on Pullman dataset

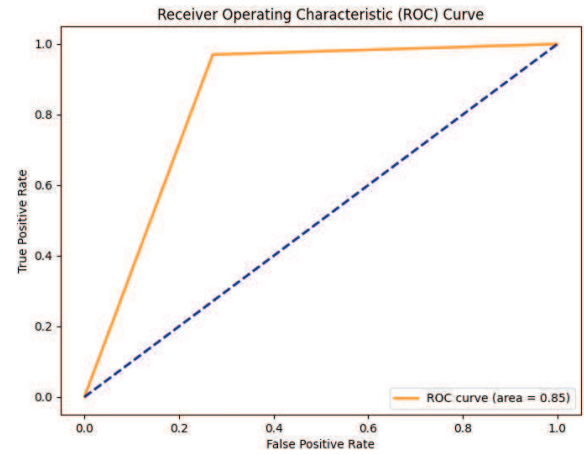


Fig. 8: ROC curve of LSTM on Hardrock dataset

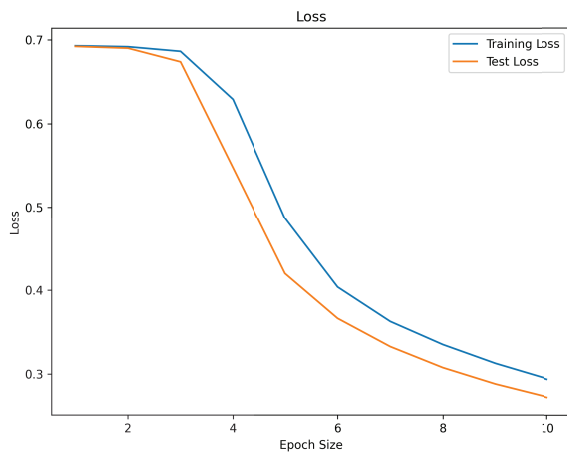


Fig. 6: Loss graphics of CNN on IMDB dataset

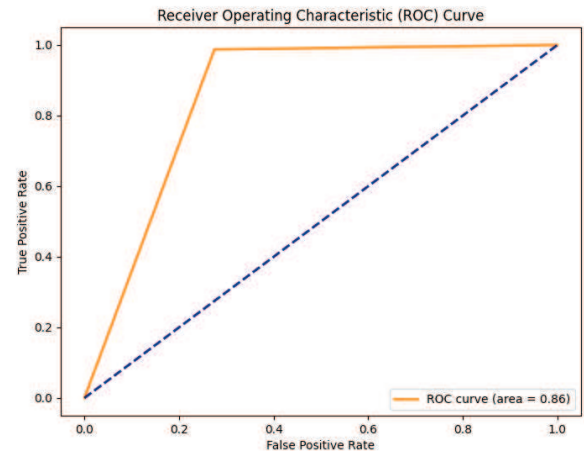


Fig. 9: ROC curve of CNN-LSTM on Ayana dataset

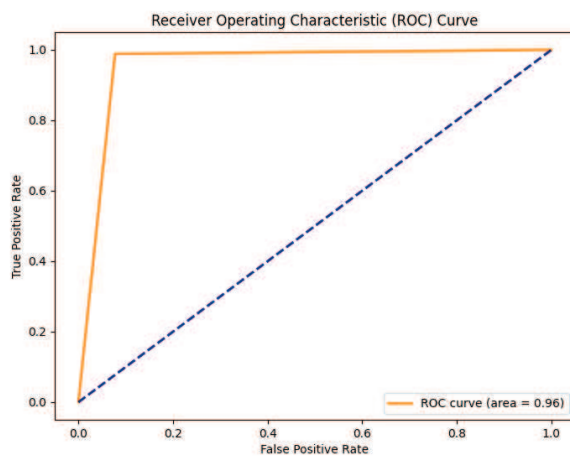


Fig. 7: ROC curve of LSTM on Padma dataset

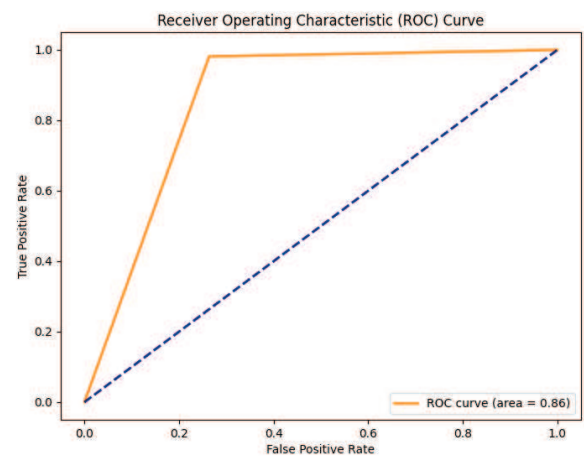


Fig. 10: ROC curve of BiLSTM on Pullman dataset

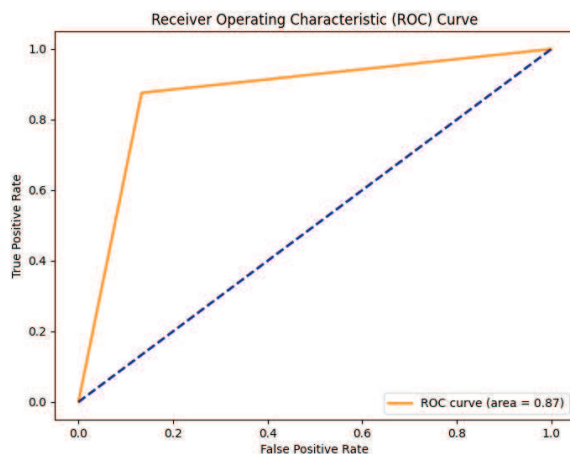


Fig. 11: ROC curve of CNN on IMDB dataset

VI. CONCLUSIONS

In conclusion, the escalating volume of online data underscores the increasing importance of sentiment analysis. This study specifically addresses the need for sentiment analysis in the realm of hotel reviews. Our investigation reveals that LSTM exhibits superior performance when applied to hotel review datasets. However, the dynamics change when dealing with a larger IMDB dataset, where the complexity of CNN results in improved performance.

This research introduces an innovative methodology powered by deep learning models aimed at evaluating customer experiences in the context of hotels. Leveraging cutting-edge technologies, our methodology employs advanced models such as CNN, LSTM, CNN-LSTM, BiLSTM, and Conv-BiLSTM. These models are strategically applied to capture the intricacies and contextual nuances embedded within customer reviews. As part of our future research direction, we intend to extend the application of transformer-based neural networks to various domains within natural language processing, evaluating their performance across diverse contexts.

However, the method presented in this paper can only categorize sentiment into positive and negative, which may not be suitable for fields demanding more nuanced sentiment distinctions. Therefore, the next step is to study the sentiment fineness classification of the text.

REFERENCES

- [1] Jain, Praphula Kumar, Rajendra Pamula, and Gautam Srivastava. "A systematic literature review on machine learning applications for consumer sentiment analysis using online reviews." *Computer science review*, 2021.
- [2] Ady, Margaret, and Donna Quadri-Felitti. "Consumer research identifies how to present travel review content for more bookings." *Hotels News Resource* 95, 2015.
- [3] Tang, Liang, Xi Wang, and Eojina Kim. "Predicting Conversion Rates in Online Hotel Bookings with Customer Reviews." *Journal of Theoretical and Applied Electronic Commerce Research*, 2022, pp. 1264-1278.
- [4] Diana Gavilan, Maria Avello, Gema Martinez-Navarro. "The influence of online ratings and reviews on hotel booking consideration." *Tourism Management*, vol 66, 2018, pp. 53-61
- [5] Meena, Gaurav, Krishna Kumar Mohbey, and Ajay Indian. "Categorizing sentiment polarities in social networks data using convolutional neural network." *SN Computer Science* 3.2, 2022.
- [6] Wang, Zhumei, Xing Su, and Zhiming Ding. "Long-term traffic prediction based on lstm encoder-decoder architecture." *IEEE Transactions on Intelligent Transportation Systems* 22.10, 2020.
- [7] Colón-Ruiz, Cristóbal, and Isabel Segura-Bedmar. "Comparing deep learning architectures for sentiment analysis on drug reviews." *Journal of Biomedical Informatics* 110, 2020.
- [8] Kapočiūtė-Dzikienė, Jurgita, Robertas Damaševičius, and Marcin Woźniak. "Sentiment analysis of lithuanian texts using traditional and deep learning approaches." *Computers* 8.1, 2019.
- [9] Muhammad, Putra Fissabil, Retno Kusumaningrum, and Adi Wibowo. "Sentiment analysis using Word2vec and long short-term memory (LSTM) for Indonesian hotel reviews." *Procedia Computer Science* 179, 2021, pp. 728-735.
- [10] Budhi, Gregorius Satia, et al. "Using machine learning to predict the sentiment of online reviews: a new framework for comparative analysis." *Archives of Computational Methods in Engineering* 28, 2021, pp. 2543-2566.
- [11] Chang, Yung-Chun, Chih-Hao Ku, and Chien-Hung Chen. "Using deep learning and visual analytics to explore hotel reviews and responses." *Tourism Management* 80, 2020.
- [12] Shaukat, Zeeshan, et al. "Sentiment analysis on IMDB using lexicon and neural networks." *SN Applied Sciences* 2, 2020, pp. 1-10.
- [13] Ketkar, Nikhil, et al. "Convolutional neural networks." *Deep Learning with Python: Learn Best Practices of Deep Learning Models with PyTorch*, 2021, pp.197-242.
- [14] Sherstinsky, Alex. "Fundamentals of recurrent neural network (RNN) and long short-term memory (LSTM) network." *Physica D: Nonlinear Phenomena* 404, 2020.
- [15] Agga, Ali, et al. "CNN-LSTM: An efficient hybrid deep learning architecture for predicting short-term photovoltaic power production." *Electric Power Systems Research* 208, 2022.
- [16] Xu, Guixian, et al. "Sentiment analysis of comment texts based on BiLSTM." *Ieee Access* 7, 2019.
- [17] Liu, Gang, and Jiabao Guo. "Bidirectional LSTM with attention mechanism and convolutional layer for text classification." *Neurocomputing* 337, 2019, pp. 325-338.
- [18] A. Maas, R. E. Daly, P. T. Pham, D. Huang, A. Y. Ng, and C. Potts, "Learning word vectors for sentiment analysis," in *Proceedings of the 49th annual meeting of the association for computational linguistics: Human language technologies*, 2011, pp. 142–150.
- [19] Moolayil, Jojo, and Jojo Moolayil. "An introduction to deep learning and keras." *Learn Keras for Deep Neural Networks: A Fast-Track Approach to Modern Deep Learning with Python*, 2019, pp. 1-16.
- [20] Tripathy, Abinash, Ankit Agrawal, and Santanu Kumar Rath. "Classification of sentiment reviews using n-gram machine learning approach." *Expert Systems with Applications* 57, 2016, pp. 117-126.
- [21] Gamal, Donia, et al. "Analysis of machine learning algorithms for opinion mining in different domains." *Machine Learning and Knowledge Extraction* 1.1, 2018, pp. 224-234.
- [22] BHassan, Sayar Ul, Jameel Ahamed, and Khaleel Ahmad. "Analytics of machine learning-based algorithms for text classification." *Sustainable Operations and Computers* 3, 2022, pp. 238-248.
- [23] Gupta, Vedika, et al. "Predicting attributes based movie success through ensemble machine learning." *Multimedia Tools and Applications*, 2022, pp. 1-30.